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### Compatibility

- The software uses some features from C++11 standard. These are integer types, smart pointers and lambda functions, among others.
- A modern compiler is required for building the library successfully. A GCC compiler of version 4.6 or later, MS Visual Studio 2010 or later, a Clang compiler of version 3.1 or later will be sufficient.



Fig. 2: Sample output after coding using the library

of the spectrum of the signal. In frequency domain, spectrum of the signal is multiplied with the filter for modifying it. As per the documentation given on the website of the software ([www.aquila-dsp.org](http://www.aquila-dsp.org)), Finite Impulse Response (FIR) and Infinite Impulse Response (IIR) filters are yet to be implemented.

**Feature extraction.** The feature extraction of the signal allows the user to reduce the signal into easily comparable forms. This is very useful in speaker-identification or speech-recognition processes.

Besides extracting power or energy of the signal, the software allows the usage of Mel Frequency Cepstral Coefficients (MFCC). Although MFCC is considered as one of the standard methods for feature extraction, its sensitivity to noise due to its dependence on the spectral form is a clear disadvantage.

In order to overcome this problem, methods that use information in the periodicity of a speech signal are used. The algorithm for finding Human Factor Cepstral Coefficients (HFCC) for feature extraction is yet to be ported from the previous to the latest version of the software.

**Dynamic time warping.** Dynamic Time Warping (DTW) algorithm was originally developed for speech recognition. Two sequences of feature vectors are time-warped and aligned. The process

continues until a perfect match between the features occurs, based on certain matrices.

### Who should use the tool

We had a quick overview over the various functionalities of the tool. The software is in a stage of continuous development, and a stable release of Aquila 3 that packs all algorithms in the previous versions ported is expected soon.

If you are a beginner trying to learn the basics of signal processing, Aquila could be an ideal tool for you. However, the library lacks advanced topics in signal processing such as wavelets and polyphase filters.

As a user at [www.experts-exchange.com](http://www.experts-exchange.com) points out, another open source software called Signal Processing Using C++ (SPUC) provides features like digital-modulation schemes, which we do not find in Aquila. So the best solution in this regard will be to use both libraries through a wrapper to combine these. **EFY**

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